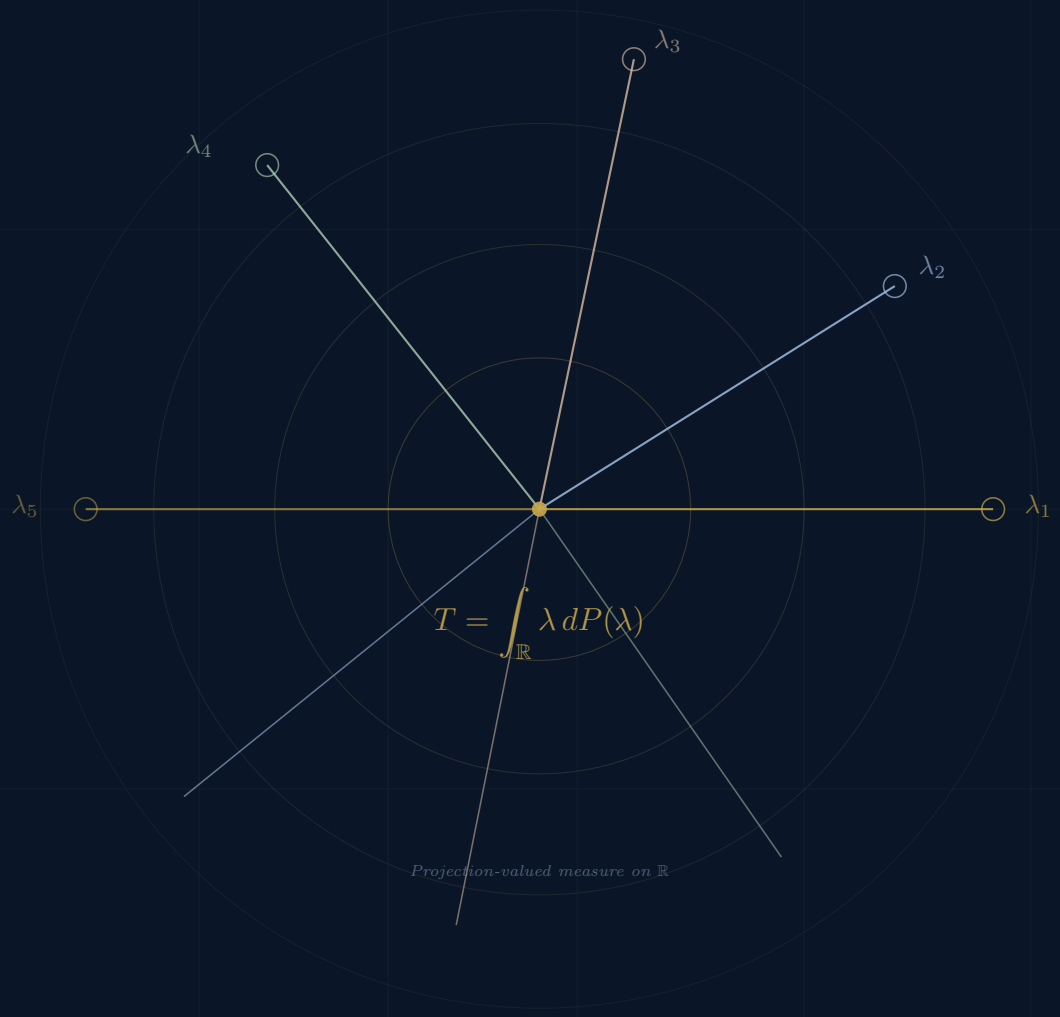
The Spectral Theorem

FOUNDATIONS · APPLICATIONS · MODERN PERSPECTIVES



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Chapter 1

Foundations and Hilbert Spaces

This chapter establishes the necessary mathematical framework for understanding the Spectral Theorem. We begin by reviewing the structure of inner product spaces, leading to the concept of completeness that defines a Hilbert space. This foundation is essential because the Spectral Theorem fundamentally relies on the properties of self-adjoint operators defined within this complete framework.

1.1 Inner Product Spaces and Norms

1.1.1 The Abstract Inner Product Space

A vector space V over the field $\mathbb{F}$ (where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) is an inner product space if there exists a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ satisfying the following axioms for all vectors $x, y, z \in V$ and scalars $a \in \mathbb{F}$:

1. **Conjugate Symmetry (or Symmetry):** $\langle x, y \rangle = \overline{\langle y, x \rangle}$. (If $\mathbb{F} = \mathbb{R}$, this simplifies to symmetry: $\langle x, y \rangle = \langle y, x \rangle$).

2. **Linearity in the First Argument:** $\langle ax + y, z \rangle = a\langle x, z \rangle + \langle y, z \rangle$.

3. **Positive Definiteness:** $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0$ if and only if $x = 0$.

Definition 1.1 (Inner Product Space)**.** A vector space equipped with an inner product is called an inner product space.

1.1.2 Induced Norms

From the inner product, we define the induced norm (or length) on V:

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

This norm induces a metric $d(x, y) = \|x - y\|$ on V, making $(V, \|\cdot\|)$ a normed vector space.

1.1.3 Orthogonality and Projections

The concept of orthogonality is central: two vectors x and y are orthogonal if $\langle x, y\rangle = 0$. The geometric interpretation of the inner product allows us to define projections onto subspaces. For any closed subspace $M \subset V$, the orthogonal projection $P_M : V \to M$ is defined such that for any $x \in V$, $P_M x$ is the unique vector in M that minimizes the distance $\|x - y\|$ for $y \in M$.

1.2 Completeness and Hilbert Spaces

1.2.1 Cauchy Sequences and Completeness

A sequence $\{x_n\}$ in an inner product space is a Cauchy sequence if for every $\epsilon > 0$, there exists an N such that for all $m, n > N$, $\|x_m - x_n\| < \epsilon$.

Definition 1.2 (Completeness)**.** An inner product space $(V, \langle\cdot,\cdot\rangle)$ is a complete space if every Cauchy sequence in V converges to a limit that is also in V.

1.2.2 The Hilbert Space Definition

A Hilbert space $\mathcal{H}$ is defined as an inner product space that is complete with respect to the norm induced by the inner product. The completeness property ensures that limits of convergent sequences remain within the space, which is crucial for analysis and solving differential equations in infinite dimensions.

1.3 Examples of Hilbert Spaces

1.3.1 Finite-Dimensional Case ($\mathbb{R}^n$ or $\mathbb{C}^n$)

The standard Euclidean spaces equipped with the standard dot product form finite-dimensional Hilbert spaces. For $x = (x_1, \dots, x_n) \in \mathbb{C}^n$, the inner product is:

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i \overline{y_i}.$$

These are trivially complete.

1.3.2 The Space ℓ^2

The space of square-summable sequences:

$$\ell^2 = \left\{ x = (x_1, x_2, \dots) \in \mathbb{C}^\infty \mid \sum_{n=1}^{\infty} |x_n|^2 < \infty \right\}.$$

The inner product is defined as:

$$\langle x, y\rangle = \sum_{n=1}^{\infty} x_n \overline{y_n}.$$

This space is a classic example of an infinite-dimensional Hilbert space.

1.4 Orthonormal Bases and Fourier Series

1.4.1 Orthonormal Sets

A set $\{e_i\}_{i=1}^{\infty} \subset \mathcal{H}$ is orthonormal if:

$$\langle e_i, e_j \rangle = \delta_{ij},$$

where δ_{ij} is the Kronecker delta.

1.4.2 Gram-Schmidt Process

Given a linearly independent set $\{v_1, v_2, \dots\}$, we can construct an orthonormal sequence using the Gram-Schmidt process:

$$u_1 = \frac{v_1}{\|v_1\|}, \quad u_2 = \frac{v_2 - \langle v_2, u_1 \rangle u_1}{\|v_2 - \langle v_2, u_1 \rangle u_1\|}, \dots$$

1.4.3 Fourier Series

In $L^2([0, 2\pi])$, the orthonormal basis is given by:

$$e_n(x) = \frac{1}{\sqrt{2\pi}} e^{inx}, \quad n \in \mathbb{Z}.$$

Any function $f \in L^2([0, 2\pi])$ can be expanded as a Fourier series:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e_n(x),$$

where $c_n = \langle f, e_n \rangle$.

1.5 Orthogonal Projections and Projection Theorems

1.5.1 Projection onto Subspaces

Let $M \subset \mathcal{H}$ be a closed subspace. For any $x \in \mathcal{H}$, the orthogonal projection $P_M(x) \in M$ satisfies:

$$x - P_M(x) \perp M.$$

1.5.2 Projection Theorem

If $M \subset \mathcal{H}$ is a closed subspace and $x \in \mathcal{H}$, then the projection $P_M(x)$ is the unique element in M that minimizes $\|x - y\|$ over all $y \in M$.

1.6 Self-Adjoint Operators and Their Properties

1.6.1 Adjoint of an Operator

For a linear operator $A : \mathcal{H} \to \mathcal{H}$, the adjoint A^* satisfies:

$$\langle Ax, y \rangle = \langle x, A^* y \rangle \quad \forall x, y \in \mathcal{H}.$$

1.6.2 Self-Adjoint Operators

An operator A is self-adjoint if $A = A^*$. This implies:

$$\|Ax\|^2 = \|x\|^2 \quad \text{for all } x \in \mathcal{H}.$$

1.6.3 Spectral Decomposition of Self-Adjoint Operators

A self-adjoint operator can be decomposed into eigenvalues and eigenvectors:

$$A = \sum_{i=1}^{\infty} \lambda_i P_i,$$

where λ_i are real eigenvalues, and P_i are orthogonal projections onto the corresponding eigenspaces.

1.7 Spectral Theorem in Finite Dimensions

1.7.1 Diagonalization of Self-Adjoint Operators

In finite dimensions, a self-adjoint operator is diagonalizable (with respect to an orthonormal basis). This leads to the spectral theorem:

$$A = UDU^*, \quad \text{where } D \text{ is diagonal and } U \text{ is unitary.}$$

1.7.2 Eigenvalues and Eigenvectors

If A has eigenvalues λ_i with corresponding orthonormal eigenvectors u_i, then:

$$A = \sum_{i=1}^{n} \lambda_i |u_i\rangle\langle u_i|.$$

1.8 Spectral Theorem in Infinite Dimensions

1.8.1 Continuous Spectrum and Discrete Spectrum

In infinite dimensions, the spectral theorem describes operators as integrals over their spectrum:

$$A = \int_{\sigma(A)} \lambda dP(\lambda),$$

where $\sigma(A)$ is the spectrum of A, and P is a projection-valued measure.

1.8.2 Normal Operators

A normal operator A satisfies $AA^* = A^*A$. The spectral theorem for normal operators generalizes the finite-dimensional case to infinite dimensions:

$$A = \int_{\sigma(A)} \lambda dP(\lambda),$$

where P is a projection-valued measure.

1.9 Conclusion

This chapter has introduced the foundational concepts of inner product spaces, Hilbert spaces, orthonormal bases, and self-adjoint operators. These ideas are essential for understanding the spectral theorem in both finite and infinite-dimensional settings, which will be explored in greater depth in subsequent chapters.

Bibliography

[1] R. B. Ash, *Real Analysis and Probability*, Academic Press, 2002.

[2] H. B. Phillips, *Linear Algebra and Its Applications*, Pearson, 2007.

[3] P. D. Lax, *Functional Analysis*, Wiley, 2002.